TYSEER NINAD

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SUMMARY

A graduate in Avionics Engineering with research experience in wireless communication systems and UAV-assisted networks, including IEEE-published work on deep learning–based signal classification and a thesis on privacy-preserving modulation recognition using distributed learning. Skilled in Python, C/C++, and MATLAB, with practical experience in system modeling, algorithm development, and performance analysis through collaborative research projects. Interested in applying machine learning and data-driven methods to communication and networked systems, particularly in UAV and IoT environments.

EDUCATION

Bachelor of Science in Avionics Engineering **JAN 2022 - JAN 2026**
Aviation and Aerospace University, Bangladesh (AAUB) | CGPA: **3.49** out of **4.00**

Higher Secondary Certificate **JUN 2018 - JAN 2021**
Notre Dame College, Dhaka | Group; Science | GPA: **5.00** out of **5.00**

Secondary School Certificate **JAN 2016 - MAY 2018**
Dhaka Government Muslim High School | Group: Science, GPA | **5.00** out of **5.00**

RESEARCH EXPERIENCE

Founder & Vice Team Lead, AAUB Rover 71 **OCT 2023 - SEP 2025**

- Led end-to-end administrative, technical, financial, and sponsorship operations across multiple workstreams and cross-functional teams, while representing AAUB and Bangladesh in internationally benchmarked events and contributing to global collaboration, technology awareness, and STEM advancement
- Planned, designed, and delivered two versions of "AAUB Rover 71," converting conceptual designs into operational systems through field-level testing and problem solving, and securing global rankings of 8th and 11th at the Qualification and Final Rounds of the Anatolian Rover Challenge 2025 in the team's first participation.
- Coordinated project outcomes through structured presentations and 4 distinct Conference Papers at multiple international flagship platforms, strengthening reporting clarity and professional communication skills.

PUBLICATIONS

[1] M. E. Rahman, A. H. Olee, B. Ahmmed, **T. Ninad**, M. S. Rahman and M. A. Hossain, "***Electro-Mechanical Design Analysis and Implementation of an Indigenous Mars Rover***," 2024 IEEE Region 10 Symposium (TENSYMP), New Delhi, India, 2024, pp. 1-6, DOI: 10.1109/TENSYMP61132.2024.10752217

[2] M. E. Rahman, **T. Ninad**, M. T. Islam, T. R. Turjo, S. Akter, M. S. Islam, M. Azam, S. M. S. Bari, "***Rover-71: An Indigenous Multi-Purpose Rover Featuring Autonomous Navigation and Control,***" 2025 7th International Conference on Electrical, Control and Instrumentation Engineering, Pattaya, Thailand, 2025, pp. 481 - 488, DOI: 10.1109/ICECIE66637.2025.11363832

[3] M. E. Rahman, A. H. Olee, **T. Ninad**, M. S. Hossain, M. A. Jamil, "***Automatic Modulation Recognition in OTFS Systems Using Optimized SE-CNN Architecture***," 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), Kushtia, Bangladesh, 2025, pp. 1 - 6, DOI: 10.1109/COMPAS67506.2025.11381657

[4] M. E. Rahman, **T. Ninad**, M. A. I. Aion, M. R. Haque, T. Ahmed, M. S. Rahman, "***LIFE-MAP: an Indigenous Biosignature Detection Protocol Using Multi-Biomolecule Based Analysis***," 2025 IEEE Region 10 Conference (TENCON), Kota Kinabalu, Malaysia, 2025, pp. 1734 - 1738, DOI: 10.1109/TENCON66050.2025.11375044

[5] A. F. Anon, M. S. Prodhan, H. P. Himel, M. T. Islam, M. R. Hasan, K. A. I. Jisan, **T. Ninad**, M. E. Rahman, M. A. Hossain, "***Design and Development of an Indigenous Modular 5-DOF Robotic Arm for Multidisciplinary Tasks***," 2025 IEEE Region 10 Conference (TENCON), Kota Kinabalu, Malaysia, 2025, pp. 391 - 395, DOI: 10.1109/TENCON66050.2025.11375029

[6] **T. Ninad**, M. E. Rahman, M. M. Hossain, M. U. Tazbid, A. Gifari, S. M. S. Bari, "***Transfer Learning for Deep Feature Extraction in Automated Rock Classification***,"
Submitted To: ICCIT 2025 ***(Peer Reviewed, Accepted for Publication)***

RESEARCH GRANTS

2024 **Developing an Integrated Communication and Autonomous Navigation System for a Remotely Operated Vehicle (ROV)**
- Principal Investigator: ***Assistant Professor Dr. S. M. Sadakatul Bari*** ; Allotted Fund: **BDT 210,000**

2024 **Design & Development of an Indigenous Modular 5-Degree Of Freedom Robotic Arm for Multidisciplinary Applications**
- Principal Investigator: ***Air Cdre (Retd) Md Afzal Hossain, BUP, ndc, psc, Ph.D.*** ; Allotted Fund: **BDT 350,000**

2024 **Design & Development of a Scientific Exploration and Environmental Analysis Module for Assessing Interplanetary Habitability**
- Principal Investigator: ***Assistant Professor Md Samin Rahman*** ; Allotted Fund: **BDT 210,000**

2023 **Designing and Developing an Indigenous Mars Rover for Inter-planetary Scientific Exploration**
- Principal Investigator: ***Air Cdre (Retd) Md Afzal Hossain, BUP, ndc, psc, Ph.D.*** ; Allotted Fund: **BDT 210,000**

THESIS

Privacy Preserving Automatic Modulation Recognition for UAV Networks using Distributed Learning

- Supervisor: ***Air Cdre Mohammed Akhter Jamil, PhD (Chairman, Department of Avionics Engineering)***
- Ex-Supervisor: ***Associate Professor Md Sakir Hossain, PhD (Department of CSE), BRAC University***
- Abstract: This thesis proposes a privacy-preserving solution for Automatic Modulation Recognition (AMR) in UAV-assisted networks, leveraging decentralized federated learning (DFL) and the OTFS Delay-Doppler domain. A novel framework is developed that optimizes both communication and privacy, while significantly enhancing AMR performance, achieving up to 100% accuracy at 6 dB SNR.

COURSE PROJECTS

Evolution of Wireless Communication and the Role of GFDM in 5G Networks [2025]

- Abstract: This course project examines Generalized Frequency Division Multiplexing (GFDM) as a flexible and spectrally efficient 5G waveform through mathematical analysis and MATLAB simulation, demonstrating improved error performance and reduced spectral leakage compared to OFDM under AWGN channels.

Quantifying Cognitive Stress: EEG Based Assessment for Aviation Professionals [2024]

- Abstract: This project applies a 1D Convolutional Neural Network to EEG signals for cognitive stress assessment in aviation contexts, employing subject independent validation and achieving 68.39 percent accuracy, highlighting the feasibility of deep learning based neurophysiological analysis for human centered applications.

Home Automation and Security System Using Logic Gates [2023]

- Abstract: Designed a low-cost home automation and security system using fundamental digital logic, integrating a binary keypad door lock, intrusion detection, conditional appliance control, and alarm functionality through modular IC-based hardware with an emphasis on reliability and scalability.

SKILLS SUMMARY

- ***Languages:*** Python, C, C++
- ***Platforms:*** Arduino, Visual Studio Code, Jupyter Notebook,
- ***Software:*** MATLAB, Simulink, TensorFlow, PyTorch, Altium, Proteus, Cadence Virtuoso
- ***Tools:*** Google Workspace, Microsoft Office Power BI, Latex, Notion, Trello
- ***Technical Expertise:*** Data Analysis, Embedded Systems Fundamentals, Simulation-based Problem Solving, System Modeling, Prototyping, Testing, Iterative Development
- ***Soft Skills:*** Project Planning & Management, Competitive Analysis, Leadership, Teamwork, Public Speaking

SCHOLARSHIPS

Awarded multiple merit-based scholarships for academic excellence, including the ***Vice Chancellor Scholarship (8th & 6th Semesters)*** and ***General Scholarship (5th & 2nd Semesters)***, with consistent high GPA performance (up to **3.89/4.00**).

CERTIFICATIONS OF CONFERENCE PRESENTATIONS

2025 **Certification of Conference Paper Presentation at IEEE ICCIT'25**
2025 **Certification of Conference Paper Presentation at IEEE ICECIE'25**
2025 **Certification of Two Conference Papers Presentation at IEEE R10 TENCON'25**
2025 **Certification of Conference Paper Presentation at IEEE COMPASS'25**
2025 **Certification of Conference Paper Presentation at IEE R10 TENSYMP'24**
2022 **Certification of Participation at IEEE CS BDC Summer Symposium'22 (Abstract)**

AWARDS & ACHIEVEMENTS

2026 **Best Thesis Poster Presentation (Undergraduate Thesis)**
2025 **Finalist (11th), Anatolian Rover Challenge 2025, Ankara, Turkey**
2025 **Certification of Excellence (Vice Team Lead), AAUB Rover 71**
2025 **Certification of Appreciation (Vice President-Avionics), AAUB AI & Robotics Club (AAIRC)**
2024 **BEAR Patron - BEAR Summit and National Semiconductor Symposium 2025**
2025 **Certification of Appreciation (Vice President - Debate & Workshop), AAUB Debating Club**
2024 **Organizer - ROBOVERSE 0.1: An Intra-University Robotics Competition, BAIRC, BSMRAAU**
2023 **Organizer – Seminar on "Trends in Aviation & Space Technology Research", BAIRC, BSMRAAU**
2022 **Certification of Participation at IAAC 2023 (Silver Honor - Final Round)**

INTERSHIPS

Intern (Technology, Planning & Implementation), Bangladesh Satellite Company Limited **FEB 2026 - PRESENT**

- Supported satellite service planning and implementation activities, gaining hands-on exposure to GEO operations, ground segment architecture, and VSAT service workflow.
- analyzed satellite communication networks, link budgets, and traffic flow to understand network performance and service architecture.
- Assisted in capacity planning, service performance monitoring, and preparation of technical reports aligned with operational and regulatory standards.

TRAINING & PROFESSIONAL EXPOSURE

Industrial Training, Biman Bangladesh Airlines **JUN 2024 - JUL 2024**

- As an Industrial Trainee, gained knowledge of airline operations, aircraft maintenance, and aviation safety protocols
- Explored and gained insights into Boeing 787-8, Boeing 787-9, Boeing 777, Boeing 737, and Dash 8—focusing on their systems, structural design, and operational capabilities
- Observed cockpit operations of the Boeing 787, including flight instruments, navigation systems, and pilot workflows
- Improved teamwork and technical communication through collaboration with aviation professionals

Industrial Visit, Air Traffic Control & Radar, Civil Aviation Authority of Bangladesh (CAAB) **NOV 2025**

- Gained practical understanding of network operations and large-scale communication infrastructure, with exposure to real-time monitoring and control environments
- Studied the working principles of radio frequency (RF) systems, signal transmission, and core hardware components used in wireless communication networks
- Analyzed network data to understand traffic flow, signal coverage, system performance, and operational efficiency in high-capacity communication systems

LEADERSHIP & VOLUNTERRING EXPERIENCE

Co-founder, Vice President (Avionics), AAUB AI & Robotics Club **JUN 2023 - SEP 2025**

- Led academic and technical engagement initiatives by organizing multiple seminars, webinars, competitions, workshops in aerospace, aviation and space robotics, with responsibility for planning, coordination, and hybrid event execution.

Vice President (Debate & Workshop), AAUB Debating Club **NOV 2024 - SEP 2025**

- Led the first membership campaign, enrolling 60+ members, and organized AAUB's inaugural Inter-Department Debate Championship (2025) by managing event planning, team registration, logistics, and on-ground execution for 10+ teams.

BEAR Patron, Bangladesh National Semiconductor Symposium '25 **JUN 2025 - JUL 2025**

- Represented the AAUB AI and Robotics Club as a BEAR Patron under the World Bank–financed EDGE Project, participating in national policy dialogues on AI, robotics, and semiconductor innovation and collaborating with academic and industry stakeholders to support youth-led technology initiatives in Bangladesh

Secretary (Organizing & Festival), Notre Dame Chess Club (NDCC) **JUN 2019 - MAR 2020**

- Coordinated major events, including the 6th Inter School-College Chess Fest, and contributed to NDCC's first magazine "Fortress" by managing sponsorships and article collection.

MEMBERSHIPS

2025 ***IEEE Student Member, Bangladesh Section***

- Membership No #101586283

RECOMMENDATIONS

Air Cdre (Retd) Md Afzal Hossain, BUP, ndc, psc, Ph.D.
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